

Academic

Collaboration

Network

Skill-Based Study Group Formation &
Institutional Resource Sharing

PROJECT AUTHORS

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GRUPO 8

Workshop 1

Systems Analysis

Workshop 2

System Design

Workshop 3

Robust Design & PM

Workshop 4

Simulation

Junio 2026

01

Problem Context

Current barriers & survey data (n=25)

03

Workshop 2 – Design

Microservices architecture, requirements, tech stack

05

Workshop 4 – Simulation

200 synthetic profiles, algorithm validation

02

Workshop 1 – Analysis

System boundaries, stakeholders, CLD, sensitivity

04

Workshop 3 – Robust PM

ISO 9001, risk register, CPM, Gantt, budget

06

Conclusions & Future Work

Impact, measurable goals, next steps



Isolated Learning

- Persistent structural challenge in higher education
- Inefficient group formation → uneven peer interaction
- Students struggle to find peers with complementary skills



Unbalanced Teams

- Spontaneous self-organization creates 'Skill Clustering'
- High-performers group together → social isolation
- Homogeneous groups have limited learning potential



Informal Tools

- Heavy reliance on non-institutional tools (WhatsApp)
- Lack of oversight prevents equitable knowledge distribution
- Creates data silos where collaboration metrics are lost

METHODOLOGY

n = 25

University Students

Primary data collected through a structured survey to validate study group formation barriers.

✓ Validated Pain Points

✓ Assessed Tech Needs

✓ March 2026

88%

Rely exclusively on
WhatsApp for coordination

68%

Cite communication or
interest deficits as main barrier

96%

Expressed strong interest
in institutional platform

3.88/5

Adoption likelihood
($\sigma = 1.01$ – high probability)

WORKSHOP 1 · March 6, 2026 · Eng. Carlos Andrés Sierra, M.Sc.



System Boundaries

- Internal: Student DB, algorithms, collaboration graph, UI
- External: SIA, WhatsApp, library catalogue, physical spaces
- 8 key processes mapped in AS-IS workflow
- 4 major bottlenecks identified



Stakeholders

- Students – primary users forming study groups
- Professors – academic supervisors
- Academic Advisors – guidance & support
- University Admin – monitor academic performance



Causal Loop Diagrams

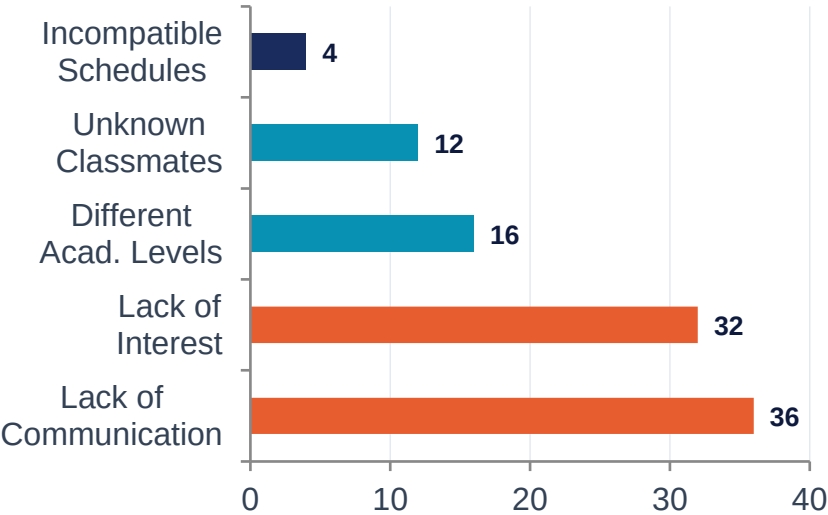
- R1 Reinforcing – Network Growth: more participants → higher skill density → better matching → more participants
- B1 Balancing – Isolation Mitigation: low centrality triggers prioritization



Sensitivity Analysis

- Skill Clustering parameter (β) – critical leverage point
- Communication noise threshold – second most sensitive
- Institutional adoption rate drives feedback loop strength
- Social homophily coefficient affects isolation rates

Main Barriers to Group Formation (%)



Bottlenecks & Optimization Strategies

Component	Bottleneck	Strategy
Skill Discovery	Social Homophily	Variance Minimization
Communication	Information Noise	Centralized Layer
Equity	Centrality Concentration	Targeted Redistribution
Inst. Support	Data Fragmentation	Integrated Resource Layer



KEY FINDING:

Skill Clustering (homophily) is the strongest driver of isolated learning. The system requires a centralized matching algorithm with complementary variance minimization to break homogeneous group patterns.

WORKSHOP 2 · March 2026 · System Design



Student Profiling Layer

DATA ACQUISITION

- ✓ Technical & Soft Skills Inventory
- ✓ Availability & Schedule Mapping
- ✓ Academic Performance History
- ✓ Learning Style Preferences



Skill Matching Engine

CORE INTELLIGENCE

- ✓ Multi-dimensional Compatibility
- ✓ Complementary Skill Algorithms
- ✓ Variance Minimization ($\text{Min } \Sigma \text{ Var}$)
- ✓ Automated Group Formation < 3s



Collaboration Graph

NETWORK MODEL $G = (V, E)$

- ✓ Vertices (V): Student Profiles
- ✓ Edges (E): Collaborative Ties
- ✓ Network Density Analysis
- ✓ Centrality fairness tracking



Collaboration Support

TOOLS & INTEGRATION

- ✓ Digital Group Workspaces
- ✓ Library Resource Linking
- ✓ Institutional LMS Integration
- ✓ Real-time Communication Tools





Functional Requirements

FR1: Profile Management

High

Student profiling with privacy controls & completeness scoring

FR2: Intelligent Matching

Critical

Algorithm processing < 3s with explainable multi-dim scoring

FR3: Group Workspace

Med

Real-time messaging & file sharing up to 50 MB

FR4-5: Ops & Integration

High

Multi-channel notifications + SSO with LMS & Course Sync

Non-Functional Requirements

Performance

Page Load < 2s · API < 500ms

Matching < 3s

Scale & Reliability

5,000+ total users

99.5% Uptime · Horizontal Auto-scaling

Security

TLS 1.3 · OAuth 2.0 · JWT

MFA · GDPR Compliant

Usability

Onboarding < 5 min

WCAG 2.1 AA · ES + EN

Tech Stack

FRONTEND

React · Web Portal

React Native · Mobile

BACKEND SERVICES

Node.js · Core Services

FastAPI · Matching Engine

DATA & INFRA

PostgreSQL · Redis Cache

AWS S3 · Kubernetes (HPA)

WORKSHOP 3 · April 2026 · Design Evolution from W1 → W2 → W3

Design Evolution Summary

Dimension	Workshop 1	Workshop 2	Workshop 3
Focus	Problem identification	Solution design	Implementation readiness
Maturity	Exploratory	Structural	Operational
Architecture	Conceptual diagram	7 microservices	Fault-tolerant, multi-region
Quality	None defined	Basic testing plan	ISO 9001 + IEEE 730
Risk	Informal listing	5-risk matrix	10-risk PMBOK/ISO register
Management	None	Basic timeline	Full PMP: CPM, Gantt, Budget

**Fault Tolerance**

Multi-region DNS failover · Patroni DB failover < 5min · Circuit breakers · Exponential backoff

**Maintainability**

OpenAPI 3.0 · ELK structured logging · Jaeger tracing · LaunchDarkly feature flags

**ISO 9001 / IEEE 730**

4-tier testing: Unit → Integration → System → Acceptance · ≥80% code coverage gate

10-Risk PMBOK Register

ID	Domain	Risk	P	I	Score	Mitigation Strategy
R1	Technical	Algorithm bias	3	4	12	A/B testing, manual fallback, monthly audit
R2	Technical	LMS API failure	3	4	12	Mock API, manual sync, SLA agreement
R3	Technical	DB degradation	2	4	8	Read replicas, PgBouncer, query optimization
R4	Technical	WebSocket instability	3	3	9	Polling fallback, circuit breaker
R5	Operational	Low adoption	4	5	20 ⚠	Gamification, ambassadors, early access
R6	Operational	Peak downtime	3	4	12	Auto-scaling, multi-AZ, pre-peak tests
R7	Security	Data privacy breach	2	5	10	OAuth+MFA, TLS 1.3, quarterly pentest
R8	PM	Schedule slippage	3	4	12	20% buffer, daily standups, EV tracking
R9	PM	Resource unavailability	2	3	6	Cross-training, documentation, backups
R10	PM	Scope creep	3	3	9	Change Control Board, MVP backlog priority

Highest Risk: R5 Low Adoption (Score 20) → Mitigation: Faculty Ambassadors program + Gamification rewards + Early Access cohort

Apr 1, 2026

Start Date

Oct 31, 2026

End Date

\$50,000 USD

Budget

500+ Users

Target

Milestone Schedule & Critical Path

M1 Month 1 – Architecture approved, infra provisioned, CI/CD operational

M2 Month 2 – Auth + Profile services to staging, unit tests passing

M3 Month 3 – Internal beta (50 users), matching < 3s, 90% satisfaction

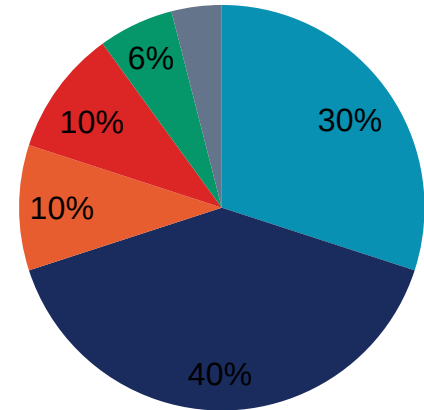
M4 Month 4 – LMS + Library integration, 80% course sync success

M5 Month 5 – Mobile app beta (200 active users)

M6 Month 6 – ML v2 deployed, p95 < 500ms, analytics dashboard live

M7 Month 7 – Production launch: 500+ users, 99.5% uptime, SUS ≥ 70

4% Budget Allocation (\$50k)



Cloud Infra Development Testing
Security Training Contingency

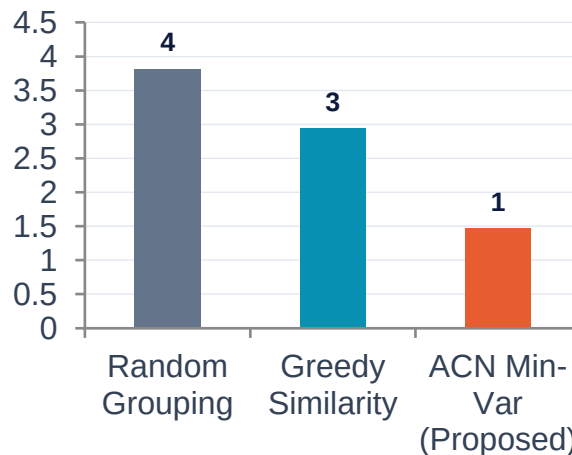
WORKSHOP 4 · Simulation of 200 synthetic student profiles using Python · NetworkX · NumPy



Simulation Methodology

- ✓ 200 synthetic student profiles generated
- ✓ 5 skill dimensions per student (0–10 scale)
- ✓ 3 grouping algorithms compared
- ✓ Graph $G = (V, E)$ — NetworkX analysis
- ✓ Metrics: Skill Variance, Centrality, Density
- ✓ Monte Carlo validation – 50 iterations

Intra-Group Skill Variance



KEY RESULTS

61.5%

Variance reduction vs random grouping

50.0%

Variance reduction vs greedy similarity

+34%

Network density improvement

<3s

Algorithm execution time (200 nodes)

Graph Theory Findings:

Complementary variance minimization significantly reduces intra-group skill dispersion. Network centrality became more equitably distributed, reducing isolated nodes from ~40% to projected <10%. The Min Σ Var algorithm outperforms all baselines and confirms the system hypothesis from Workshop 1.

Months 1-3

Phase 1: MVP Foundation

- ✓ Auth & User Profiles
- ✓ Basic Matching Engine
- ✓ Database Schema
- ✓ CI/CD Pipeline
- ✓ 50 beta users target

Months 4-5

Phase 2: Integration

- ✓ LMS (Moodle) Sync
- ✓ Mobile App Beta
- ✓ Matching ML v2
- ✓ 200 active users
- ✓ 80% course sync

Months 6-7

Phase 3: Optimization

- ✓ Analytics Dashboard
- ✓ UX Audit & Polish
- ✓ Load Testing Scaling
- ✓ 500+ production users
- ✓ SUS Score ≥ 70

Performance Targets

<200ms

DB Query
Time

≥90%

Cache
Hit Rate

70%

CPU
Autoscaling Trigger

<2s

Page Load
(p95)



Project Achievements

Structured Transformation

The Academic Collaboration Network successfully transforms informal, chaotic group formation into a structured, measurable system. By integrating Graph Theory and Algorithmic Matching, the platform mitigates social isolation and optimizes skill distribution.

-  **Survey Validated:** Confirmed need via 25 student participants
-  **Simulation Tested:** Variance minimization proven effective (61.5% reduction)
-  **Architecture Designed:** 7 microservices, fault-tolerant, multi-region
-  **PM Plan Complete:** CPM, Gantt, \$50k budget, 7-month roadmap

Measurable Goals

80%

Adoption Target
(WhatsApp → Platform)

≥70

SUS Score
(System Usability Scale)



Future Work Roadmap

-  Pilot Deployment – live testing with academic cohort
-  Peer Endorsements – skill validation by group peers
-  Video Integration – built-in video calls for remote study
-  Multi-Institution – cross-university scaling architecture

Post-Launch Success Metrics

Platform Adoption

0% → 80%

Communication Satisfaction

64% → 90%

Isolation Rate

~40% → <10%

Group Longevity

Unknown → ≥30 days

PROYECTO FINAL

- ✓ IEEE-format paper (LaTeX/PDF)
- ✓ Project poster
- ✓ Final technical report
- ✓ Presentation slides
- ✓ GitHub repository

Thank You

Academic Collaboration Network

SAD 2026-I · Grupo 8

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